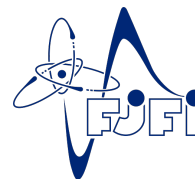




02YMECH/02YMECHZ Mechanics



Lecturer: İskender Yalçinkaya, Ph.D.
Email: iskender.yalcinkaya@fjfi.cvut.cz
Web: <https://people.fjfi.cvut.cz/yakcuisk/index.html>
Office: B-509*
Office Hours: By appointment

*From the main staircase of the Břehová building, go up to the 4th floor, then take the stairs on your left toward the roof and ring the bell marked Q3.

Course Description

02YMECH is a winter semester course for first-year bachelor students covering the basics of mechanics and some advanced methods, such as solving equations of motion for mechanical systems like a particle under constant force, a harmonic oscillator, the central force problem, and rigid bodies.

Each week, there will be four 50-minute lectures and two 50-minute exercise session. Lectures will cover theoretical concepts following the weekly schedule, while exercise sessions will focus on solving problems related to the lecture material. The instruction language is English.

02YMECHZ refers to the final exam for 02YMECH. The students who pass 02YMECH will be eligible to take 02YMECHZ. Please see the exam section for more details.

Prerequisite(s)

Knowledge of basic differentiation, integration, and trigonometry would be useful but are not necessary.

Credits: 4 (02YMECH) + 2 (02YMECHZ)

Reference Textbooks

1. Morin, D. J., *Introduction to Classical Mechanics: With Problems and Solutions*, Cambridge University Press, 2008.
2. Taylor, J. R., *Classical Mechanics*, University Science Books, 2013.
3. Thornton, S. T., Marion, J. B., - *Classical Dynamics of Particles and Systems*, 3rd Edition, Brooks/Cole, 2003.
4. Halliday, D., Resnick, R., Walker, J., *Fundamentals of Physics*, 10th Edition (Extended), Wiley, 2013.

Course Objectives

1. To learn the basics of mechanics, and solve simple equations of motion.
2. To understand the general framework for constructing and operating physical models.

Homeworks

Each week, students will be given homework on the topics covered in class. The purpose of homework is that students can work on their own and familiarize themselves with the subject. Homework will not be returned and will not be graded. Solutions to the homework assignments will be provided one week before the upcoming exam date.

Exams

There will be **two midterm exams** and **one final exam** (02YMECHZ). Each exam will contain **four questions**, with each question worth **one point**. Students who score at least **50% of the total points** from the midterms will be awarded the **Z (Zápočet) grade**, meaning they will pass 02YMECH and receive 4 credits. To qualify for the final exam, students must obtain this Z grade from the midterms. The final exam consists of a **written and an oral component**:

- Only students who score at least **50% on the written part** will proceed to the oral exam.
- The **final letter grade** is primarily determined by the written exam score. However:
 - If the oral exam score is **above 70%** the final grade is increased by one level.
 - If the oral exam score is **below 30%**, the final grade is decreased by one level.
 - If the oral exam score is **from 30% to 70%**, the final grade remains the same.
- Attendance at the oral exam is **mandatory** for students who pass the written part.
- The structure of the oral part will be as follows: Each student will choose a random number from 1 to 16, corresponding to one of the sixteen topics listed at the end of this syllabus (please see “Topics for the Oral Part of the Final exam”). The students will then have 30 minutes to prepare a presentation on the chosen topic on a blank paper, followed by a 5 to 10-minute oral examination based on that topic. The students do not need to provide all the derivations (unless otherwise stated) but should be able to explain the main concepts and equations related to the topic.

The dates of the exams will be announced in the second week of the semester in KOS. The students are supposed to register for the exams in KOS. The students are required to bring their student ID cards to the exams.

Letter Grade Distribution

[3.6 – 4.0] : A = Excellent	[2.4 – 2.8) : D = Average
[3.2 – 3.6) : B = Very Good	[2.0 – 2.4) : E = Sufficient
[2.8 – 3.2) : C = Good	[0.0 – 2.0) : F = Insufficient

Makeup Exams

If a student fails to obtain at least 50% of the total points from the midterm exams required to qualify for the final exam, they are eligible to take a makeup exam. The makeup exam will be scheduled before the official examination period. The structure and grading of the makeup exam will be identical to the midterms and will cover all the topics from the first two midterms. After taking the makeup exam, the student must achieve at least 50% of the total points from the combined score of the midterms and the makeup exam in order to be eligible to attend the final exam.

Retaking the Course

If a student fails the course for the first time, they may retake it in the following academic year. The student is required to attend all classes and take the exams again. If the student fails the retake, they must obtain special permission from the study department to retake the course one final time.

Attendance and Absences

- Attendance is strongly recommended but not obligatory.
- Students are responsible for all missed work, regardless of the reason for absence. It is also the absentee's responsibility to get all missing notes or materials.

Tentative Course Outline

The weekly coverage might change as it depends on the progress of the class.

Chapter	Week
Mathematical preliminaries	1
Kinematics	2
Dynamics	3
Rotation	4
Conservation theorems	5,6
Small oscillations	7,8
Central-force motion	9,10
Dynamics of system of particles	11
Motion in a non-inertial reference frame	12
Dynamics of rigid bodies	13

Week	Content
Week 1	Scalars, scalar functions, cartesian and polar coordinates, derivatives, integrals, introduction to first- and second-order homogenous differential equations, vectors, vector operations.
Week 2	Reference frames, position, displacement, velocity, acceleration, speed, average speed, momentum, basic types of motion and their superposition.
Week 3	Inertial and non-inertial reference frames, Newton's laws of motion, equation of motion for a particle under a constant force, inclined plane, tension, drag force, friction, freefall and terminal speed, projectile motion.
Week 4	Angular velocity, torque, angular momentum, rotational inertia of a system of particles, Newton's 2nd law in rotational form, uniform circular motion.
Week 5	Conservation of linear and angular momentum, work and kinetic energy, conservative forces and potential energy.
Week 6	Conservation of total energy, total energy as an equation of motion, equilibrium, impulse, power.
Week 7	Simple harmonic oscillator, simple pendulum, phase diagrams for the oscillative behavior.
Week 8	Damped oscillations, driven oscillations, resonance.
Week 9	Definition of central force and its properties, equations of motion for central force problem, orbits, centrifugal energy, effective potential.
Week 10	Two-body problem, planetary motion (cyclic, elliptic, parabolic, and hyperbolic orbits, eccentricity, and total energy), Kepler's laws of planetary motion.
Week 11	Center of mass, reconsideration of basic mechanical quantities, dynamics and conservation laws for system of particles, elastic and inelastic collisions.
Week 12	Centrifugal and coriolis forces, motion relative to the Earth, oblateness of Earth, cyclonic motion in atmosphere.
Week 13	Yo-yo problem, physical pendulum, inertia tensor, Euler's equations, force-free motion of a symmetric top.

Topics for the Oral Part of the Final Exam

1. How would you describe a circular motion in terms of a complex number? If you wanted the radius of the circular trajectory to decrease exponentially with time, how would you modify your description? What would be the connection of this description to damped oscillations?
2. Derive work-energy theorem step-by-step starting from the definition of work and Newton's second law and explain its meaning.
3. Define conservative force and potential energy by using the definition of work and explain why a conservative force can be expressed as the negative gradient of potential energy.
4. Show mathematically step-by-step in detail why the total mechanical energy is conserved for a particle moving under a conservative force.
5. Use the conservation of mechanical energy as an equation of motion to derive the time dependence of position for a simple harmonic oscillator in one dimension.
6. Derive step-by-step of total kinetic energy for discrete and continuous mass distributions starting from the definition of kinetic energy for a single particle and describe of how we define moment of inertia on the way.
7. Describe projectile motion and derive expressions for time of flight, maximum height, range and trajectory equation.
8. If the potential energy of a mechanical system can be described as an analytical function and if we are interested in small oscillations around the equilibrium point, show step-by-step how we can approximate the potential energy function using Taylor series expansion and derive the equation of motion in differential equation for for small oscillations. Do not solve it but memorize the general solution in terms of trigonometric functions.
9. Starting from Hooke's law, derive the equation of motion for a simple harmonic oscillator and solve it step-by-step to get the general solution. Explain the physical meaning of each term in the solution.
10. Starting from the expression $x(t) = x_0 \cos(\omega_0 t - \delta)$, show the form of the trajectories followed by a simple harmonic oscillator in phase space (the phase diagram). Describe all relevant features of the phase diagram in terms of physical parameters of the system, such as total energy, amplitude, frequency, and mass. How would the phase diagram change if we introduce damping into the system?
11. Write down the equation of motion for a damped harmonic oscillator under the influence of a damping force proportional to velocity. Solve the equation step-by-step and obtain the general solution in exponential form. Explain briefly how different types of damping based on the relationship between the damping coefficient and the natural frequency of the system lead to different forms of the motion (underdamping, overdamping, and critical damping).
12. Write down and solve step-by-step the equation of motion for a driven damped harmonic oscillator under the influence of an external sinusoidal driving force to derive the expression for the amplitude and phase difference of steady-state oscillations. Explain the concept of resonance and how it manifests in this system.

13. Define central force and describe its three main properties: No torque, motion in a plane, equal areas in equal times. Show mathematically step-by-step how each property can be derived from the definition of central force.
14. Describe how two-body problem can be reduced to an equivalent one-body problem. Derive the equation of motion for the equivalent one-body problem step-by-step.
15. Write down the effective potential for inverse-square central force and draw its graph. Explain how different types of orbits (circular, elliptical, parabolic, hyperbolic) can be understood in terms of the effective potential and total energy of the system.
16. Write down the effective force in a non-inertial reference frame including all non-inertial terms. Explain the physical meaning of each term in detail. Describe the cyclonic motion in the atmosphere in terms of these non-inertial forces and explain why the wind blows counterclockwise (clockwise) around low-pressure areas in the northern (southern) hemisphere.